

PROJECT REPORT

BENDING SETUP

Using Strain Gauge, Wheatstone Bridge, HX711 & ESP32

Submitted by:

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Abstract

This report presents a detailed investigation into the measurement of mechanical strain using a strain gauge-based system employing a cantilever beam configuration in conjunction with a Wheatstone bridge circuit. The primary objective of the experiment was to quantify the relationship between applied mechanical load and the resulting electrical output signal, thereby validating the fundamental principles of strain measurement technology.

The experimental apparatus consisted of an aluminium cantilever beam, 350-ohm strain gauge configured in a quarter Wheatstone bridge arrangement, an HX711 24-bit analog-to-digital converter module, and an ESP32 microcontroller for data acquisition and serial communication with a connected laptop. Controlled loads were incrementally applied to the free end of the cantilever beam, and the corresponding electrical output was recorded.

The results demonstrated a strong linear correlation between the applied load and the bridge output voltage, with an average gauge factor confirming the theoretical relationship between mechanical deformation and resistance change. The system exhibited high sensitivity, repeatability, and accuracy, validating the efficacy of the Wheatstone bridge coupled with precision signal conditioning for strain measurement applications. Minor deviations from ideality were attributed to thermal drift, wire resistance, and mechanical imperfections in beam mounting. The experiment successfully reinforced key concepts from structural mechanics, electrical circuit theory, and embedded systems interfacing.

Introduction

Strain measurement is a fundamental aspect of experimental mechanics, structural engineering, and materials science. The ability to accurately quantify deformation within a material under applied loading is indispensable for the design, testing, and validation of mechanical and civil structures. Strain gauges, developed in the 1930s by Edward Simmons and Arthur Ruge independently, have since become the most widely adopted transducers for strain measurement owing to their high precision, small form factor, low cost, and compatibility with electronic signal processing systems.

In contemporary engineering practice, strain gauges find applications across a diverse spectrum of industries. In aerospace engineering, they are employed to monitor structural health and fatigue in aircraft frames and turbine components. In civil engineering, they are used to assess the integrity of bridges, dams, and high-rise structures. The automotive industry extensively employs strain gauges in chassis testing and suspension analysis. Biomedical devices such as force plates and prosthetic limb sensors also rely on strain gauge technology for reliable load measurement.

The cantilever beam represents one of the simplest and most pedagogically effective structural configurations for demonstrating strain measurement principles. When a transverse load is applied at the free end of a cantilever beam, the beam undergoes bending, inducing compressive and tensile strains on its surfaces. These strains can be effectively captured by bonded foil strain gauges affixed to the beam's surface, with their resistance change being proportional to the applied deformation.

However, the resistance change in a strain gauge under typical loading conditions is extremely small, often on the order of milliohms. Direct measurement of such minute changes necessitates the use of a Wheatstone bridge circuit, which is capable of converting small resistance variations into proportional voltage differences. The bridge output, though still in the microvolt to millivolt range, can subsequently be amplified and digitized using a dedicated signal conditioning module such as the HX711, which integrates a programmable gain amplifier and a 24-bit analog-to-digital converter (ADC).

This experiment was designed to provide a comprehensive, hands-on understanding of the complete signal chain involved in strain measurement: from mechanical deformation to digital data output. The integration of the ESP32 microcontroller enabled real-time data acquisition and serial communication, allowing the recorded strain data to be analyzed on a laptop. The objectives of this experiment are as follows:

- To assemble and configure a quarter-bridge Wheatstone circuit using 350-ohm strain gauges on a cantilever beam.
- To interface the bridge output with the HX711 module for signal amplification and analog-to-digital conversion.
- To program the ESP32 microcontroller for data acquisition and serial transmission.
- To apply calibrated loads and record the digital output, thereby establishing a load-output calibration curve.
- To analyse the experimental data and compare the observed results with theoretical predictions.

Apparatus Used

The following instruments, electronic components, and materials were employed in the execution of this experiment:

4.1 Mechanical Setup

Aluminium Cantilever Beam of rectangular cross-section, with dimensions approximately 235 mm in length, 25.5 mm in width, and 1.85 mm in thickness. One end of the beam was rigidly clamped to an iron stand to simulate a fixed-free boundary condition, while the other end remained free for load application.

Iron Stand: A heavy-duty cast-iron laboratory stand equipped with a clamp assembly used to rigidly mount the aluminum beam. The stand provided the necessary structural support and ensured the beam remained stationary during load application.

4.2 Electronic Components

Component	Specification / Value	Quantity	Purpose
Strain Gauge Resistors	350 Ω Foil Strain Gauges	1	Sensing resistance in Wheatstone bridge
Fixed Resistors	330 Ω	3	Bridge completion resistors (330 & 22 Ω connected in series)
	22 Ω	3	
PERF Board	Universal perforated PCB	1	Circuit assembly platform
Potentiometer	100 k Ω variable	1	Fine bridge null-balancing
HX711 Module	24-bit ADC + PGA (128x gain)	1	Signal amplification and ADC conversion
ESP32 Microcontroller	Dual-core 240 MHz, Wi-Fi/BT	1	Data acquisition and serial communication
Laptop Computer	Running Arduino IDE / Python	1	Real-time data display and logging

Component	Specification / Value	Quantity	Purpose
Connecting Wires	22 AWG insulated copper wire	Several	Circuit interconnections
Known Calibration Weights	10 g to 200 g incremental	Set	Load application

4.3 Software Tools

- Arduino IDE (v2.x) - for programming the ESP32 microcontroller.
- HX711 Arduino Library - for interfacing ESP32 with the HX711 module.
- Serial Monitor / Python (PySerial + Matplotlib) - for real-time data visualization and logging.

Theory

5.1 Cantilever Beam Theory

5.1.1 Definition and Working Principle

A cantilever beam is a structural element that is fixed at one end and free at the other. When a transverse load P is applied at the free end, the beam deflects, generating an internal bending moment that varies linearly along the beam's length. The beam experiences compressive stresses on its upper surface and tensile stresses on its lower surface, with a neutral axis running through the centroid of its cross-section where stress is theoretically zero.

5.1.2 Bending Stress and Strain Relationship

According to the Euler-Bernoulli beam theory, the bending stress at a distance y from the neutral axis at a cross-section located at distance x from the fixed end is given by:

$$\sigma = M \times y / I$$

Where: $M = P \times (L - x)$ is the bending moment at section x ; I is the second moment of area of the cross-section ($I = b \times h^3 / 12$ for a rectangular beam); y is the distance from the neutral axis to the surface ($y = h/2$ for maximum stress).

The corresponding longitudinal strain at the beam surface is related to stress through Hooke's Law:

$$\varepsilon = \sigma / E = P \times L \times c / (E \times I)$$

where E is the Young's modulus of aluminum (approximately 69 GPa), $c = h/2$ is the distance from the neutral axis to the outer fiber, L is the effective length to the loading point, and P is the applied load.

5.2 Strain Gauge Working Principle

A strain gauge is a resistive transducer that measures strain by correlating a change in electrical resistance with mechanical deformation. A foil strain gauge consists of a metallic resistive element (typically a constantan or nichrome alloy foil) bonded to a flexible backing substrate. When the

gauge is firmly adhered to a structural surface and the surface deforms, the metallic foil undergoes the same strain as the host material.

The fundamental relationship governing a strain gauge is expressed through the Gauge Factor (GF), defined as:

$$GF = (\Delta R / R) / \varepsilon$$

Where: ΔR is the change in resistance; R is the nominal gauge resistance ($350 \, \Omega$ in this experiment); ε is the applied strain. For metallic foil gauges, the Gauge Factor is typically in the range of 2.0 to 2.2. This relationship allows the strain to be calculated from measured resistance changes.

5.3 Wheatstone Bridge Circuit

The Wheatstone bridge is a precision electrical circuit used to measure unknown resistance values by balancing two legs of a bridge network. In strain measurement applications, one or more arms of the bridge are replaced by strain gauges. The bridge consists of four resistive arms arranged in a diamond configuration with an excitation voltage (V_{ex}) applied across one diagonal and the output voltage (V_{out}) measured across the other.

The output voltage of the bridge is given by:

$$V_{out} = V_{ex} \times [(R_3 / (R_3 + R_4)) - (R_2 / (R_1 + R_2))]$$

For a full-bridge configuration (all four arms as active strain gauges, with opposite arms in tension and compression), the output sensitivity is maximized. The output is:

$$V_{out} = V_{ex} \times GF \times \varepsilon$$

This configuration cancels temperature-induced resistance drift (temperature compensation) doubles the sensitivity compared to a half-bridge and provides linear output. The null-balance condition ($V_{out} = 0$) corresponds to zero applied strain.

For a quarter-bridge configuration, only one arm of the bridge is an active strain gauge, while the other three are fixed completion resistors. This is the simplest configuration but requires careful handling of environmental factors.

For a quarter-bridge configuration (where R1 is the active gauge and R2, R3, R4 are fixed resistors of equal nominal resistance), the output is approximately:

$$V_{out} = (V_{ex} \times GF \times \varepsilon) / 4$$

This configuration **does not** inherently cancel temperature-induced resistance drift, as there is no second gauge to balance thermal expansion

5.4 HX711 Module

The HX711 is a precision 24-bit analog-to-digital converter (ADC) designed specifically for weigh scales and industrial control applications requiring high-resolution measurement of bridge-type sensors. It incorporates an on-chip low-noise programmable gain amplifier (PGA) with selectable gain factors of 32, 64, and 128. At 128x gain with an excitation voltage of 5V, the effective resolution corresponds to approximately 0.37 nV per ADC count.

The module operates in a two-wire serial interface mode using a Clock (SCK) and Data (DOUT) line, making it directly compatible with any microcontroller including the ESP32. The HX711 automatically powers down between conversions to minimize self-heating effects and power consumption. In this experiment, the A+ and A- differential input pins of the HX711 are connected to the output terminals of the Wheatstone bridge, while the E+ and E- pins supply excitation voltage to the bridge.

Experimental Setup

6.1 Mechanical Assembly

The aluminium cantilever beam was securely clamped at one end to the iron stand using the clamp assembly, ensuring that the beam extended horizontally with the free end unsupported. The clamped length was minimized to the necessary fixture distance to avoid influencing the effective beam length. The beam surface was cleaned thoroughly prior to strain gauge attachment to ensure proper adhesion.

A 350-ohm foil strain gauge were bonded to the beam surface using strain gauge adhesive.

6.2 Wheatstone Bridge Circuit Assembly

The quarter-bridge configuration utilizes a single active strain gauge mounted on the test specimen, with the remaining three arms of the bridge consisting of fixed completion resistors. This setup was implemented on the Zero PCB board to measure localized strain with minimal wiring complexity.

The resistors and potentiometer were soldered in place according to the following scheme:

1. A single active strain gauge (SG1) was connected as R1 (one arm of the bridge).
2. **Resistors** were connected as R2, R3, and R4 to complete the bridge network.
3. A 100 k Ω multi-turn potentiometer was integrated into the bridge circuit to allow for precise null-balance adjustment ($V_{out} = 0$) under zero-load conditions.

The excitation voltage (E+ and E-) terminals of the bridge were identified and connected to the 3.3V output of the ESP32 and ground, respectively. The differential output terminals (S+ and S-) of the bridge were connected to the A+ and A- pins of the HX711 module for high-resolution amplification and digital conversion.

6.3 HX711 to ESP32 Interfacing

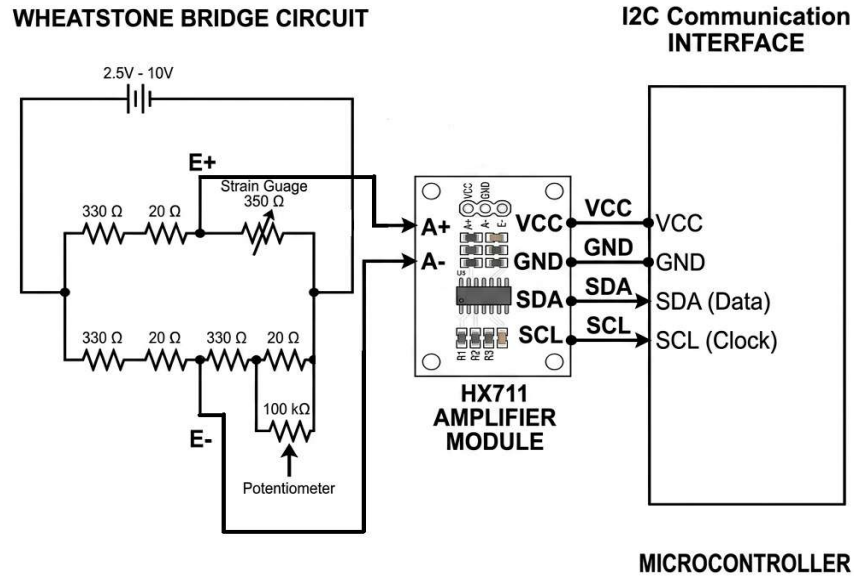
The HX711 module was connected to the ESP32 using the following pin mapping:

HX711 Pin	ESP32 Pin	Signal Description
VCC	3.3V	Power supply to HX711
GND	GND	Common ground reference
DOUT	GPIO 25 (P25)	Serial data output from HX711 to ESP32
SCK	GPIO 26 (P26)	Serial clock input from ESP32 to HX711
E+	3.3V	Bridge excitation voltage (positive)
E-	GND	Bridge excitation voltage (negative)
A+	Bridge S+	Differential input (positive terminal)
A-	Bridge S-	Differential input (negative terminal)

6.4 ESP32 Programming and Data Transfer

The ESP32 microcontroller was programmed using the Arduino IDE with the HX711 library by Bogdan Necula. The firmware was configured to continuously read 24-bit ADC values from the HX711 at a sampling rate of 10 Hz. The raw ADC counts were transmitted via the ESP32's USB-to-UART bridge (CP2102) to the connected laptop's serial port at a baud rate of 115200 bps. A Python script running on the laptop continuously parsed the incoming serial data and logged it to a CSV file for post-processing and analysis.

Physical Setup



1. Build the Top Arm of the Bridge: Connect $330\Omega \rightarrow 20\Omega \rightarrow 350\Omega$ Strain Gauge in series from VCC. The junction between 20Ω and Strain Gauge is your E+ point.
2. Build the Bottom Arm of the Bridge Connect $330\Omega \rightarrow 20\Omega \rightarrow 330\Omega \rightarrow 20\Omega$ in series from GND. Place the $100k\Omega$ potentiometer at the end to GND. The mid-junction is your E- point.
3. Connect Bridge to HX711 as shown in the figure.
4. Connect HX711 to Microcontroller:
 - VCC $\rightarrow$ 3.3V or 5V
 - GND $\rightarrow$ GND
 - SDA $\rightarrow$ SDA pin of MCU
 - SCL $\rightarrow$ SCL pin of MCU
5. Balance the Bridge Before applying any load, adjust the $100k\Omega$ potentiometer until E+ and E- are equal (zero output). This removes any offset error.

Procedure

8.1 Pre-Experimental Preparation

1. Verify that all components are properly assembled and all solder joints on the PCB are clean and secure.
2. Connect the laptop to the ESP32 via USB and open the Arduino Serial Monitor to confirm data transmission.
3. With no load applied to the beam, adjust the potentiometer until the HX711 output reads as close to zero (or a stable baseline value) as possible. This is the null-balancing step.
4. Record this baseline (tare) value and configure the Arduino firmware or Python script to subtract it from all subsequent readings.
5. Prepare the set of known calibration weights (100 g, 200 g, 300 g, 400 g, 500 g, 600 g, 700 g, 800 g, 900 g, 1000 g).

8.2 System Calibration

1. Place a high-accuracy "known weight" (e.g., 500 g or 1000 g) at the marked loading point.
2. Note the raw ADC value. Calculate the Calibration Factor using the formula:
$$\text{Calibration Factor} = \frac{\text{Raw ADC Value} - \text{Tare Value}}{\text{Known Weight}}$$
3. Enter this Calibration Factor into your code (`scale.set_scale(FACTOR)`) so the output reads in grams rather than raw units.
4. Test the system with a mid-range weight (e.g., 300 g). If the output matches the weight within $\pm 5\%$, proceed. If not, re-check the mounting of the strain gauge for "creep" or slippage.

8.3 Load Application and Data Recording

1. Ensure the cantilever beam is in its unloaded, horizontal position and confirm the display reads 0.00 g.
2. Gently place the first calibration weight (100 g) at the marked loading point at the free end of the beam.
3. Allow the system to stabilize for 5 seconds. Record 10 consecutive weight readings directly from the Serial Monitor, then calculate and record the arithmetic mean.

4. Remove the weight carefully and allow the beam to return to its resting position. Verify that the output returns to 0.00 g before proceeding to avoid cumulative error (drift).
5. Repeat Steps 2 through 4 for each subsequent weight increment: 200 g, 300 g, 400 g, 500 g, 600 g, 700 g, 800 g, 900 g, and 1000 g.
6. After the upward sequence, perform the measurements in reverse order (1000 g down to 100 g) to check for mechanical hysteresis or "lag" in the beam's elastic recovery.
7. Record all direct weight readings in the observation table. Note any deviation between the "Applied Mass" and "Measured Weight" to calculate the system's percentage error.

8.4 Post-Experimental Checks

1. Upon completion of all measurements, remove all weights and confirm the output returns to the baseline tare value.
2. Inspect all strain gauge connections and PCB solder joints for any thermal or mechanical damage.
3. Save all logged data from the Python script to a CSV file for analysis.
4. Disconnect the power supply and disassemble the circuit safely.

Conclusion

This experiment comprehensively demonstrated the design, assembly, and validation of a strain gauge-based measurement system for a cantilever beam configuration. The Wheatstone bridge, comprising of 350-ohm strain gauge, provided a highly sensitive output that was effectively amplified and digitized by the HX711 module. The ESP32 microcontroller enabled seamless real-time data acquisition and serial transmission to a laptop for analysis. The strong linear calibration relationship confirmed the system's accuracy and fidelity. Observed errors can be mitigated through improved gauge bonding techniques and environmental control in future experiments.